Daniel B. Cooney

Program in Applied and Computational Mathematics
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RESEARCH INTERESTS

Mathematical Interests: Dynamical Systems, Applied PDEs, Network and Graphon Modeling
Biological Interests: Theoretical Ecology, Evolutionary Dynamics, Evolutionary Game Theory, Multilevel Selection, Collective Behavior, Pattern Formation

EMPLOYMENT (UPCOMING)

Simons Postdoctoral Fellow, Center for Mathematical Biology
University of Pennsylvania, Philadelphia, PA, USA — Starting August 2020
Departments of Mathematics and Biology
Postdoctoral Mentors: Yoichiro Mori and Joshua Plotkin

EDUCATION

Ph.D., Applied and Computational Mathematics — September 2015 to June 2020
Princeton University, Princeton, NJ, USA
Advisor: Simon A. Levin
Dissertation: "PDE Models for Multilevel Selection: The Evolution of Cooperation and the Shadow of Lower-Level Selection"
M.A. awarded in September 2017

B.A., Cum Laude in Mathematics — August 2010 to May 2014
Harvard University, Cambridge, MA, USA
Concentration in Mathematics with Secondary Field in Economics
Senior Thesis: "Rock-Paper-Scissors: An Application of Evolutionary Game Theory to Electoral Politics," advised by Clifford H. Taubes

PUBLICATIONS

4. **D. Cooney** (2020). Analysis of Multilevel Replicator Dynamics for General Two-Strategy Social Dilemmas. *Bulletin of Mathematical Biology*, *82(6)*, 1-72.
3. **D. Cooney** (2019). The Replicator Dynamics for Multilevel Selection in Evolutionary Games. *Journal of Mathematical Biology*, *79(1)*, 101-154.
2. **D. Cooney**, B. Allen, C. Veller (2016). Assortment and the Evolution of Cooperation in a Moran Process with Exponential Fitness. *Journal of Theoretical Biology*, *409*, 38-46.
1. V. Wong, **D. Cooney**, & Y. Bar-Yam (2016). Beyond Contact Tracing: Community-Based Early Detection for Ebola Response. *PLoS Currents: Outbreaks*.

PREPRINTS

3. R. Carmona, **D. Cooney**, C. Graves, M. Laurière (2019). Stochastic Graphon Games: I. The Static Case. *arXiv preprint arXiv:1911.10664*.

2. **D. Cooney** (2019). Assortment and Reciprocity Mechanisms for Promotion of Cooperation in a Model of Multilevel Selection. *arXiv preprint arXiv:1910.06933*.

1. **D. Cooney**, F. Prieto-Castrillo, & Y. Bar-Yam (2016), Analysis of Infection-Recovery Epidemic Models for Membership Dynamics of Online Social Networks. *arXiv preprint arXiv:1608.07870*.

HONORS & AWARDS

- Best Poster Award, 6th Butler Memorial Conference on Differential Equations and Population Biology, University of Alberta, Edmonton (July 2018)
- Princeton University C.V. Starr Fellowship (2015-2016 academic year)
- NSF Graduate Research Fellowship Honorable Mention in Mathematical Biology (2016) and in Applied Mathematics (2015)
- Harvard Program for Research in Science and Engineering (PRISE) Fellow (Summer 2013)
- Harvard University Certificate of Distinction in Teaching (Fall 2012)

TEACHING EXPERIENCE

Assistant in Instruction Fall 2016 — Present
Princeton University
Led problem sessions(*) and graded homework and exams($^\#$) for

- EEB 533: Topics in Ecology: Theoretical Ecology$^{*,\#}$ (Fall 2018)
- MAE 305/MAT 391: Mathematics in Engineering I$^{*,\#}$ (Fall 2018)
- APC 199: Math Alive$^\#$ (Spring 2018)
- MAT 203: Advanced Vector Calculus$^{*,\#}$ (Fall 2017)
- MAT 103: Calculus 1$^\#$ (Fall 2016, Fall 2018)
- MAT 175: Multivariable Calc for Econ/Life Sci$^{*,\#}$ (Fall 2016)

Course Assistant Fall 2011 — Fall 2013
Harvard University
Led weekly problem sessions, held office hours, graded homework and exams for

- Math 116: Convexity and Optimization (Fall 2012, 2013)
- Computer Science 20: Discrete Mathematics for Computer Science (Spring 2013)
- Fun & Games with Discrete Mathematics (January 2012, 2013)
- Math 23a,b: Linear Algebra and Real Analysis I & II (Fall 2011, Spring 2012)

RESEARCH TALKS

- "Assortment and Reciprocity Mechanisms for Promotion of Cooperation in a Model of Multilevel Selection"
  - AMS Special Session on "Mathematical Methods for Ecology and Evolution in Structured Populations", held online due to COVID-19 (March 21, 2020)
- "PDE Models for Multilevel Selection: The Ghost of Lower-Level Selection and Transitions in Biological Complexity"
  - Mathematical Biology Seminar, Brandeis University (online; April 8, 2020)
  - Northeast Regional Conference on Complex Systems(online; April 1, 2020)
  - Collective Information Processing, Humboldt University of Berlin (March 6th, 2020)
  - Probability Seminar, Indiana University Bloomington (January 31, 2020)
  - Mathematical Biology Seminar, University of Pennsylvania (January 24, 2020)
  - AMS Western Section Meeting, University of California, Riverside (November 9, 2019)
  - ICMA VII: Mathematical Modeling and Analysis of Populations in Biological Systems, Arizona State University, Tempe (October 12, 2019)

  - Levin Fest: A Symposium at the Intersection of Mathematics and Biology, University of Victoria (June 15, 2019)
- "Ecology and Infectious Disease on Large Metapopulation Graphs"
  - Joint Mathematics Meetings, Denver, CO (January 18, 2020)
  - Flash Talk: Dynamics Days US, Hartford, CT (January 3, 2020)
- "The Replicator Dynamics for Multilevel Selection in Evolutionary Games"
  - SIAM Conference on Applications of Dynamical Systems, Snowbird, UT (May 23, 2019)
  - AMS Southeastern Section Meeting, Auburn University (March 17, 2019)
  - APS March Meeting, Boston, MA (March 4, 2019)
  - Joint Mathematics Meetings, Baltimore, MD (January 19, 2019)
  - Dynamics Days US, Evanston, IL (January 5, 2019)
  - AMS Eastern Sectional Meeting, University of Delaware (September 30, 2018)
  - Workshop on Evolutionary Models of Structured Populations, Max Planck Institute for Evolutionary Biology, Plön, Germany (September 5, 2018)
  - Applied Mathematics Seminar, University of Victoria (July 19, 2018)
- "Emergence of Efficient Extraction in Social-Ecological Models for Fisheries"
  - YSSP Final Colloquium, IIASA, Laxenburg, Austria (August 25, 2017)
- "Evolutionary Game Theory with Selection at Multiple Scales"
  - Finger Lakes Probability Seminar, Syracuse University (April 21, 2017)
- 14 talks at Princeton Theoretical Ecology Lab Tea and 6 talks at Princeton PACM Graduate Student Seminar

POSTER PRESENTATIONS

- "Ecology and Pattern Formation on Large Metapopulation Graphs"
  - SIAM Workshop on Network Sciences (online; July 2020)
- "The Replicator Dynamics for Multilevel Selection in Evolutionary Games"
  - SMB Annual Meeting, Montreal, QC (July 23, 2019)
  - Conference on the Evolution of Complex Life, Georgia Tech, Atlanta (May 15, 2019)
  - LMS School on PDEs in Mathematical Biology, ICMS, Edinburgh, UK (April 29, 2019)
  - KI-Net Young Researchers Workshop on Kinetic Descriptions in Theory and Applications, University of Maryland, College Park (October 22, 2018)
  - 6th Butler Memorial Conference on Differential Equations and Population Biology, University of Alberta, Edmonton (July 24, 2018)
- "Local and Global Stability Results for Infectious-Recovery Epidemic Models"
  - XXXV Dynamics Days US, Durham, NC (January 9, 2016)

WORKSHOPS & SUMMER SCHOOLS

- LMS Research School on "PDEs in Mathematical Biology", ICMS, Edinburgh, UK (4/2019)
- Global Biosocial Complexity Initiative Winter School on "Agent-Based Modeling of Social-Ecological Systems", Arizona State University (1/2019)
- MBI Current Topics Workshop on"Collective Behavior and Emergent Phenomena in Biology", Ohio State University (9/2018)
- AMS Mathematics Research Community on "Agent-Based Modeling in Biological and Social Systems", West Greenwich, RI (6/2018)
- "Winter Workshop on Complex Systems", University of Utrecht, Netherlands (2/2018)

- KI-Net Workshop on "Mean-field modeling and multiscale methods for complex physical and biological systems", UC Santa Barbara (9/2016)
- KI-Net Summer School on "Quantum and Kinetic Theory for Complex Systems", UC Santa Barbara (6/2016)
- AIM workshop on "Mathematics Inspired by Immuno-Epidemiology", San Jose, CA (8/2015)
- IPAM summer school on "Games and Contracts for Cyber-Physical Security", UCLA (7/2015)

TRAVEL GRANTS

- SIAM Student Travel Grant, SIAM Workshop on Network Science (July 2020)
- Collaborate@ICERM Funding to participate in small working group on "Mathematical Models of Pedestrian Movement in Large Lecture Halls" (June 2020)
- AMS Graduate Student Travel Grant, Joint Mathematics Meetings (January 2020)
- SIAM Student Travel Grant, SIAM Conference on Applied Dynamical Systems (May 2019)
- AMS Graduate Student Travel Grant, AMS Fall Eastern Sectional Meeting (September 2018)

ACADEMIC SERVICE

- Organizer
  - AMS Special Session on "Mathematical Methods for Ecology and Evolution in Structured Populations", AMS Eastern Sectional Meeting (held online; March 2020)
  - Princeton NSF Research Training Grant Seminar Series on "Mathematical Methods for Water Problems" (Fall 2018 to present)
  - Princeton Theoretical Ecology Lab Tea (Fall 2017)
- Committee Member
  - Rights & Rules Committee, Council of the Princeton University Community (2019-2020)
  - Princeton Graduate Student Government Representative for PACM (4/2017-5/2020)
  - Princeton University Ad Hoc Committee on Campus Recreation (9/2017-4/2020)
- Referee for *PNAS*, *Mathematical Biosciences*, and *Acta Mathematica Scientia*
- Academic Mentor
  - Mentor/Coach for final presentations of undergraduate applied mathematics senior independent work (Spring 2020)
  - Princeton Mathematics Mentoring Möbius Program (9/2017-1/2020)
  - Co-mentored Princeton undergraduate Alice Lin on research project "Spatial property rights as a management strategy for common-pool resources" (Summer 2017)
  - Non-Resident Tutor at Harvard College's Leverett House (2014-2015)

SHORT-TERM RESEARCH POSITIONS

Visiting Graduate Researcher — June 2017 — August 2017
International Institute for Applied Systems Analysis — Laxenberg, Austria
Participant in Young Scientists Summer Program (YSSP)

Researcher — September 2014 — August 2015
New England Complex Systems Institute — Cambridge, MA

Undergraduate Researcher — January 2013 — August 2013
Harvard Program for Evolutionary Dynamics — Cambridge, MA

COMPUTER SKILLS

Proficient with Python, Mathematica, MATLAB, LaTeX; Familiar with C

PROFESSIONAL SOCIETIES

Society for Indisutrial and Applied Mathematics (SIAM): 2015 to present
American Mathematics Society (AMS): 2018 to present
American Physical Society (APS): 2018 to present
Society for Mathematical Biology (SMB): 2020 to present